# Introduction & Perception

## Introduction to Computer Graphics - Lecture 01

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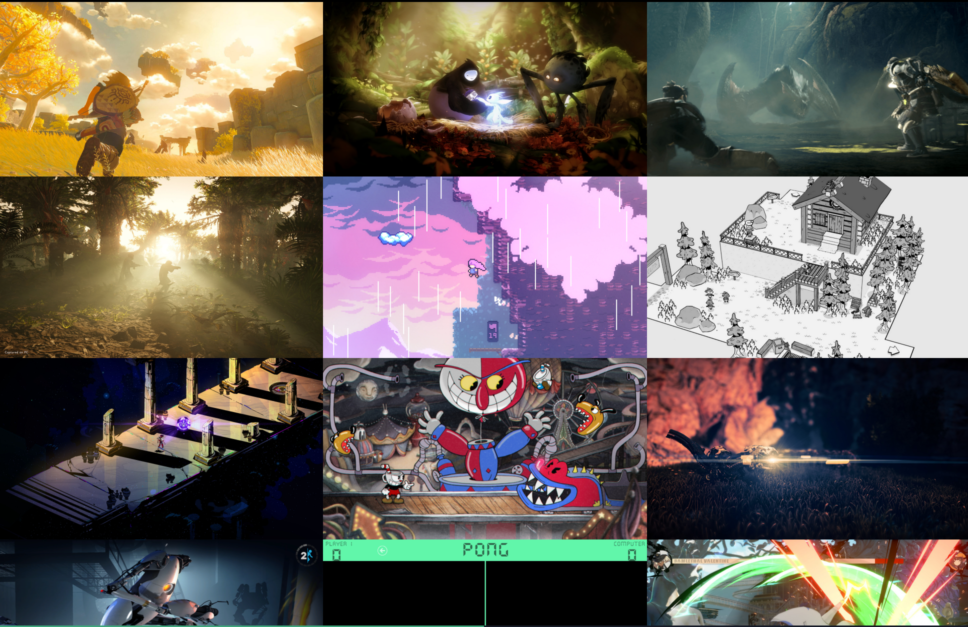
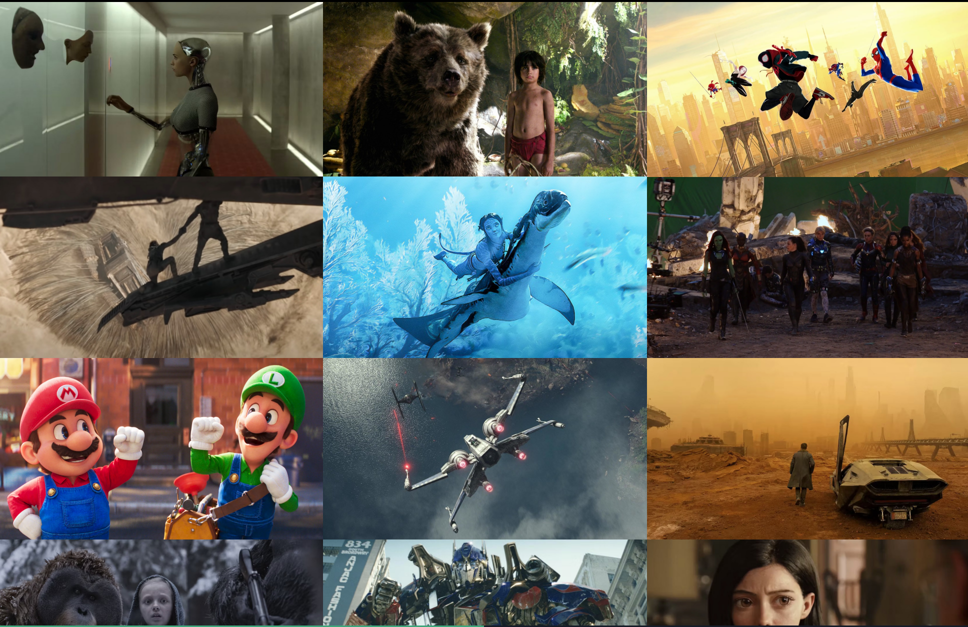
# Computer Graphics

## The creation and manipulation of images with a computer

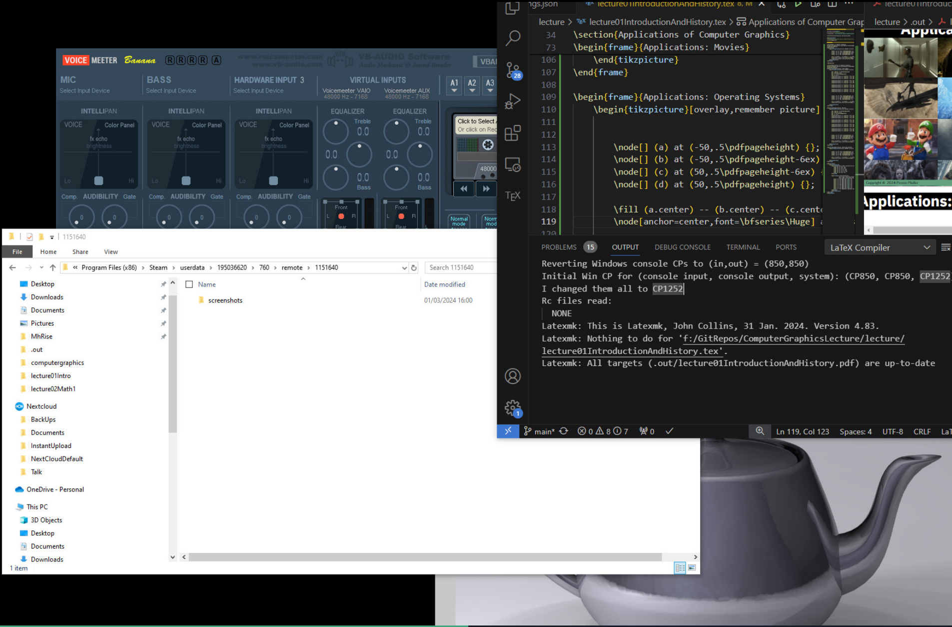
# Applications: Video Games



# Applications: Movies



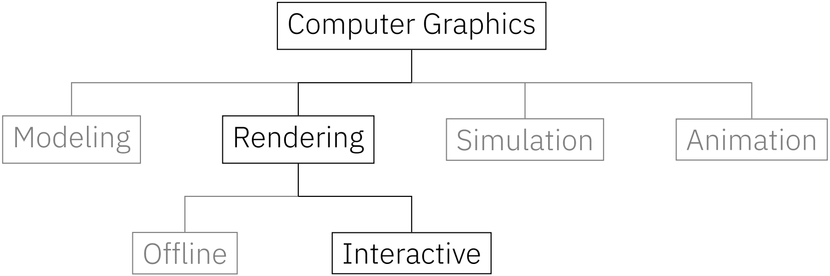
# Applications: GUI



# Areas of Computer Graphics

- Computer Graphics
  - Modeling
  - Rendering
    - Offline
    - Interactive
  - Simulation
  - Animation

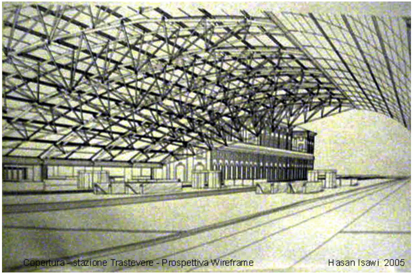
# Areas of Computer Graphics



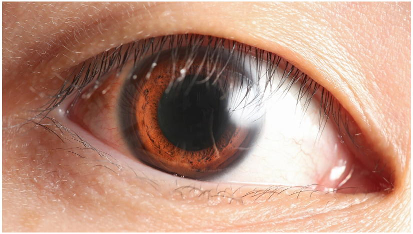

In this lecture we focus on the realtime-rendering aspect of Computer Graphics.

# Rendering Question

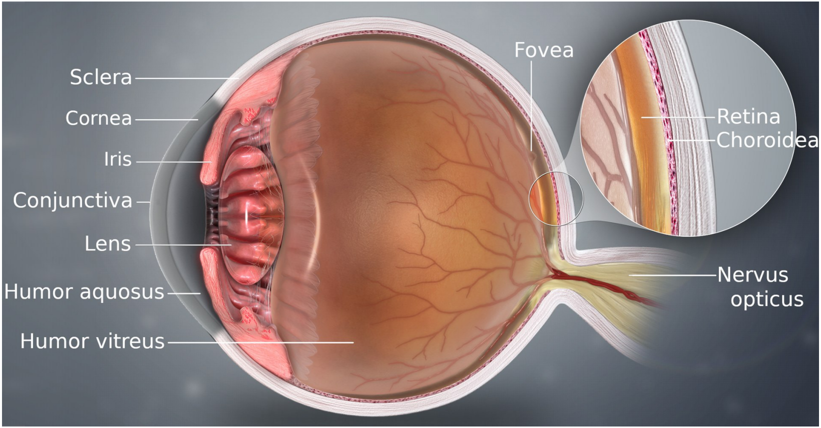
Given:
A mathematical representation of a Scene



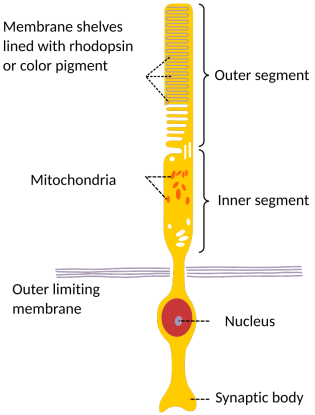

Wanted:
Something to look at (like a picture)

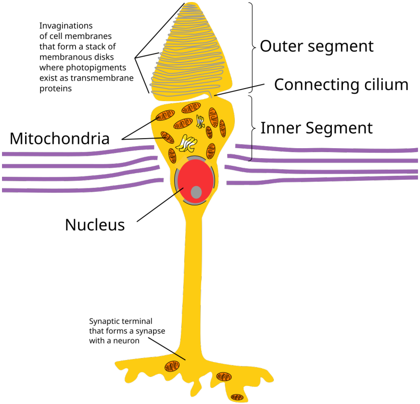


# The Human Eye



# The Human Eye



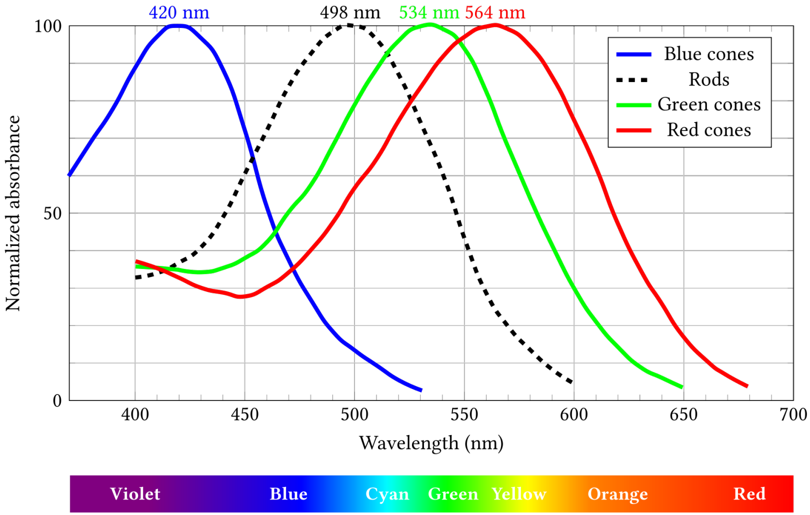




Rod Cell: Primarily perceive light intensity

Cone Cell: Primarily perceve colors (blue, green & red)

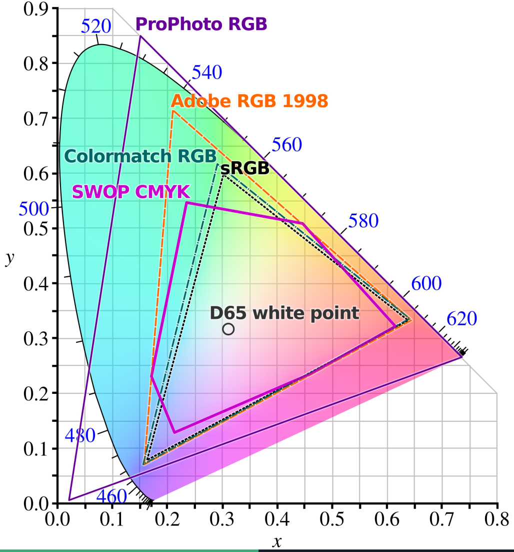
# Color Spaces

The Human Eye is not equally sensory to all colors:

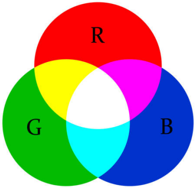


# Color Spaces

Our Devices try to replicate the natural light sensations:

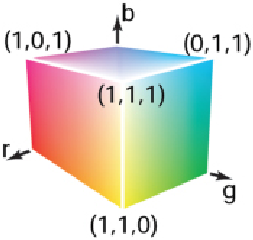


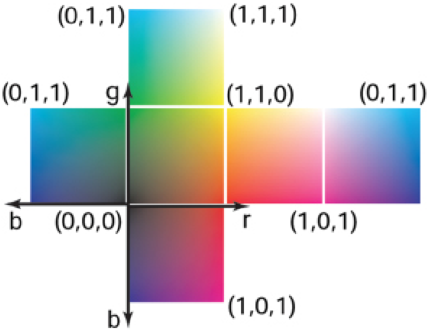
# Color Formats: RGB



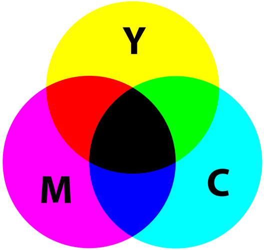

RGB is an **additive** color format with three components:

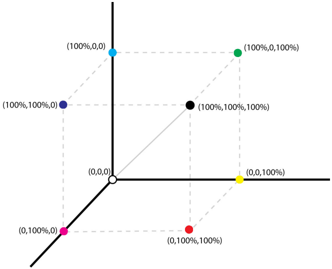
- Red
- Green
- Blue



# Color Formats: CMY




CMY is an **subtractive** color format with three components:

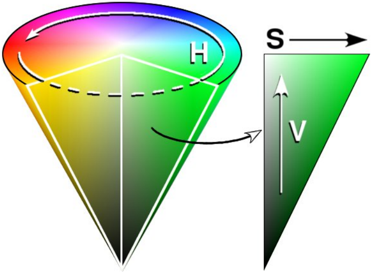
- Cyan
- Magenta
- Yellow

CMY is mainly used by printers.
CMY and RGB can be converted:

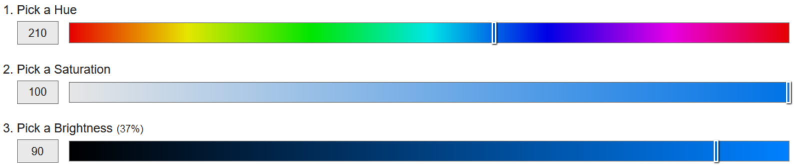
$$RGB = 1 - CMY$$

$$CMY = 1 - RGB$$

# Color - HSV



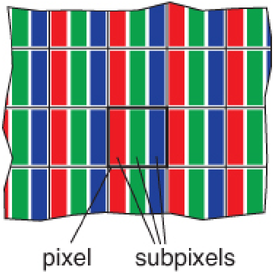
H
S
V

1. Pick a Hue
210
2. Pick a Saturation
100
3. Pick a Brightness (37%)
90

# Pixel[1]

Most images are presented on a kind of *raster display*.




Usually each pixel contains:

- Red Subpixel
- Green Subpixel
- Blue Subpixel

Common Raster Devices:

Output:
- Display
- Projector

Hardcopy:
- ink-jet printer

Input:
- Digital camera
- Scanner

[1] Pixel is short for "picture element"